Bijit Sahu

Backend & AI Engineer | Distributed Systems | LLM / RAG | Observability

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Summary

AI-focused Backend Software Engineer with **3+ years** of experience building scalable **distributed systems**, intelligent backend platforms, and high-throughput microservices for large-scale consumer applications. Experienced in implementing **LLM-powered workflows**, evidence pipelines, and **anomaly detection systems with multiple stages**. Strong foundation in **Java, Spring Boot, Kafka, microservices, and RAG applications**. Seeking a **backend AI engineering or forward deployed role** owning AI systems end to end.

Education

Year	Degree/Certificate	Institute	CGPA
2023	B.Tech + M.Tech (Dual Degree)	NIT, Rourkela	8.05

Technical Skills

- **Languages:** Java, JavaScript, TypeScript, C++, SQL
- **Backend:** Spring Boot, Microservices, REST APIs, Event Driven Architecture, Distributed Systems, System Design
- **AI / LLM:** LLM Integration, RAG, Semantic and Hybrid Search, Prompt Engineering, Agentic AI, MCP Servers, Spring AI
- **Data and Messaging:** Kafka, MySQL, PostgreSQL, Prometheus, Redis, Elasticsearch
- **Cloud and DevOps:** Docker, Kubernetes, AWS, Keycloak, Git
- **Frontend:** React, Next.js, HTML, CSS
- **Core Concepts:** System Design, Data Structures & Algorithms, OOP, Design Patterns, DBMS, Operating Systems, Concurrency

Experience

Senior Software Engineer – Rakuten Symphony | Oct 2025 – Present

- **AutoRCA – LLM-powered root cause analysis**
  - Built an RCA pipeline that gathers alarms, logs, metrics, pod state and service topology, and posts a cited first diagnosis to the incident ticket, cutting time to first diagnosis from **90–120 minutes to 15–30 minutes**.
  - Handled tickets with **1,000+ alarms** by grouping near duplicates into up to **20 investigated clusters**; under load, 10 concurrent workers processed **50 tickets in ~23 minutes**, **~36% faster** than sequential.
  - Improved answer reliability with structured outputs, a NO_DATA refusal path, per answer scope lines and stored prompts for auditability; in a review of 9 tickets, **7 of 9 initial diagnoses were usable (5 exact, 2 partial)**, with the 2 misses driving the next prompt and guardrail iteration.
- **Anomaly Detection**
  - Replaced static thresholds with a staged detection pipeline combining STL/MAD baselines that account for seasonality, correlation across signals, statistical filtering and Mahalanobis verification. Reduced **~900 raw flags to 2–3 verified alarms** every 15 minutes (**~300x noise reduction**).
  - Achieved **100% recall** on the multivariate detection stage, identifying all **60 injected faults** with **zero false positives**.
- **Distributed Tracing**
  - Built trace exploration APIs and integrated OpenTelemetry instrumented trace data that turns a single trace ID into a waterfall across services, covering **10–15 microservices**, replacing log correlation across teams with **one lookup**.
  - Developed trace visualizations with attributes at the span level, execution timelines, error details and stack traces, along with filtering and search by service in the Service Monitoring dashboard.

Software Engineer 1 – Candescent (formerly NCR Corporation) | Jul 2023 – Oct 2025

- **iSAT – Intelligent Solution and Automation Testing**
  - Automated bank transaction testing and feature certification workflows for **500+ banks**, managing **10,000+ test parameters**.
  - Delivered three modules: Archives (centralized parameter management replacing Excel), Orchestrator (VM parameter injection and automation with shell scripts) and Analytics (job status, execution progress, reporting).
  - Cut testing time by **~90%**, shortened CAB certification from **1 month to 3–4 days** and reduced manual reporting effort by **~70%**.

Projects

- **Tixolve – AI Knowledge and Support Platform** | GitHub | Aug 2025 – Present
  - Built an AI support platform for document search, Q&A, and automated query resolution using **Java 21, Spring Boot, and Spring AI**.
  - Implemented **hybrid retrieval** with PostgreSQL/pgvector and **RAG responses with citations**, enabling users to verify answers against source documents.
  - Integrated conversational support with **ticket creation through Kafka**, status tracking, and live notifications using WebSockets.

Achievements

- Solved **800+** algorithm and data structure problems on LeetCode, CodeChef and HackerRank.
- LeetCode rating **1710+** (**top 14% globally**); CodeChef **4 star**.
- Google Kickstart 2020: **rank 575**.